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Enhancing nutritional status prediction through attention-based deep learning and explainable AI

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ABSTRACT

Accurate and interpretable prediction of child malnutrition remains a critical challenge, as existing AI models often lack the transparency needed for clinical adoption. This study introduces a deep learning framework enhanced with Multi-Head Attention (MHA) for nutritional status prediction, offering a novel contribution through the first direct head-to-head comparison of CNN-MHA and LSTM-MHA to evaluate the effectiveness of spatial feature learning versus sequential dependency modeling in structured anthropometric tabular data. Our framework integrates advanced preprocessing techniques, feature selection, and Explainable AI (SHAP), enabling clinically aligned and transparent predictions. Experimental results on a 9605-sample dataset reveal that CNN-MHA achieves superior performance (99.08 % accuracy) compared to LSTM-MHA (98.91 %), confirming that spatial modeling is better suited for this dataset type. SHAP-based feature attribution further validates WHO-standard z-scores as the most influential predictors, enhancing model credibility for clinical application. Additionally, the study introduces an IoT-enabled anthropometric data acquisition system, enhancing real-time monitoring and scalability. This research represents a significant methodological advancement in nutritional status prediction, addressing key gaps in feature prioritization, accuracy, and interpretability. By bridging the gap between high-accuracy AI and clinical transparency, this study advances AI-driven nutritional monitoring and offers a scalable, explainable framework for public health interventions. Future research should explore multi-modal data integration to further enhance generalizability and real-world applicability.

1. Introduction

Child malnutrition remains a significant public health challenge, particularly in low- and middle-income countries, where early detection and timely intervention are critical to improving long-term health outcomes. In 2022, approximately 148.1 million children under the age of five experienced stunting, 45.0 million suffered from wasting, and 37.0 million were classified as overweight based on their height [1]. These statistics underscore the urgent need for scalable and effective strategies to address the nutritional needs of vulnerable populations. Traditional methods of assessing nutritional status rely heavily on manual anthropometric data collection, which is often labour-intensive and subject to errors and inconsistencies. The lack of timely and accurate data collection limits the effectiveness of intervention programs, such as the National Program for Accelerating Stunting Prevention. Therefore, leveraging advanced technologies to improve the accuracy and efficiency of children nutritional prediction is imperative for addressing this persistent issue.

The integration of Machine Learning (ML) and the Internet of Things (IoT) presents a ground-breaking approach to overcoming these challenges. This synergy leverages IoT-enabled devices for real-time data acquisition and ML algorithms for advanced predictive analysis, creating an efficient and scalable framework to monitor and address nutritional health issues. Moreover, both ML and IoT have proven

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effective in analysing public health data [2–5], making them well-suited for the characteristics of children nutrition related data [6,7]. IoT-enabled anthropometric devices integrate sensor-based measurement tools with real-time transmission technologies, ensuring high precision and automation in nutritional assessments. The system employs Rotary Encoder Sensors, Load Cell Sensors, and a Microcontroller Processing Unit to capture and transmit key anthropometric indicators, reducing human error. This combination of IoT and ML not only enhances the reliability and scalability of nutritional monitoring but also paves the way for proactive, data-driven interventions to address malnutrition more effectively [8].

Deep learning models, such as CNN and LSTM, provide a robust foundation for analyzing high-dimensional, complex data [9,10]. CNN excels at extracting spatial relationships between features [11,12], while LSTM is particularly suited for capturing complex patterns in sequential data [9,13,14], such as child anthropometric data of children. The nutritional dataset used in this study consists of structured tabular data containing anthropometric and demographic features, which pose unique challenges for deep learning models. Traditional CNN and LSTM architectures struggle to fully capture the complex relationships within this type of data. CNN, which excels at detecting spatial patterns, assumes that input features have a structured grid-like format, making it less effective for modeling interactions between non-adjacent features such as weight-for-age, height-for-age, and upper arm circumference. This limitation prevents CNN from fully learning the dependencies among nutritional indicators. On the other hand, LSTM is designed for sequential data, where time-dependent patterns play a crucial role. However, in the case of nutritional status prediction, feature relationships are not strictly sequential, meaning that treating all features as a time-series may introduce unnecessary constraints. Moreover, LSTM applies equal weighting to all past time steps, which may lead to misinterpretation of key anthropometric trends. These challenges highlight the need for a more flexible approach that can dynamically learn feature importance rather than relying on predefined spatial or sequential assumptions.

To overcome these limitations, this study proposes Multi-Head Attention (MHA) as a solution, enabling the model to learn cross-feature dependencies dynamically and adaptively prioritize the most influential features. Unlike CNN, which applies fixed convolutional filters, MHA flexibly assigns importance to different feature interactions, allowing the model to understand how multiple anthropometric indicators jointly affect nutritional status. Similarly, MHA enhances LSTM by selectively focusing on key past observations rather than treating all historical data equally, ensuring that the most relevant growth trends are captured effectively. Additionally, MHA operates in parallel rather than sequentially, improving computational efficiency while enhancing interpretability through attention weight visualization. By integrating MHA separately into CNN and LSTM models, this study aims to evaluate and compare their effectiveness in feature representation and classification accuracy, providing insights into whether spatial or sequential attention mechanisms are more suited for nutritional risk prediction.

While MHA enhances feature selection and improves predictive accuracy, deep learning models still operate as black boxes, limiting their usability in healthcare decision-making. This lack of transparency is particularly concerning when predicting nutritional status, where clear justification of feature importance is essential for clinical [15,16]. For instance, where anthropometric indicators such as weight, height, upper arm circumference, and Z-scores play a crucial role, the inability of traditional models to explain feature contributions limits their applicability in real-world healthcare decision-making. Explainable AI (XAI) addresses this limitation by enhancing model transparency through feature attribution techniques such as SHapley Additive Explanations (SHAP), which highlight the most influential factors driving predictions. XAI provides actionable insights into feature importance [17,18], enabling stakeholders to interpret and trust the model's predictions. Therefore, by integrating Explainable AI (XAI) into CNN and LSTM architectures, this study ensures that predictions align with scientifically validated indicators, particularly World Health Organization (WHO)-standard Z-scores for weight-for-age, height-for-age, and weight-for-height. These Z-scores serve as critical benchmarks in nutritional assessment, reinforcing that anthropometric features play a dominant role in classification [19], By leveraging XAI techniques, this approach ensures that AI-driven predictions are not only accurate but also interpretable, aligning with clinical best practices and providing actionable insights for health professionals to support evidence-based nutritional interventions.

Building on the need for both high predictive accuracy and model interpretability, this study integrates Multi-Head Attention (MHA) and Explainable AI (XAI) into CNN and LSTM architectures with the following contributions:

1. Enhancement of accuracy through MHA: Demonstrates that MHA enables dynamic feature prioritization, capturing complex cross-feature dependencies in structured anthropometric tabular data. Unlike traditional CNN and LSTM models, which struggle with non-adjacent feature relationships, MHA enhances feature selection and representation, leading to improved predictive performance.
2. Head-to-Head comparative analysis of CNN with MHA vs. LSTM with MHA: This study conducts the first direct evaluation of spatial feature learning vs. sequential dependency modeling for structured anthropometric data, identifying the optimal deep learning approach for nutritional status prediction
3. Integration of XAI for model transparency and WHO-standard alignment: Incorporates SHAP to ensure alignment with WHO-standard Z-scores. This integration provides feature attribution insights, enhancing model interpretability, reliability, and usability for health professionals and policymakers in guiding evidence-based nutritional interventions.

By achieving a balance between predictive accuracy and interpretability, this framework effectively addresses key technical challenges while aligning with global initiatives such as the Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger) and SDG 3 (Good Health and Well-being). Through actionable insights and real-time predictions, the proposed approach has the potential to significantly improve child nutrition outcomes.

2. Related work

The integration of IoT and AI in healthcare has significantly enhanced nutritional monitoring and patient care, particularly in managing nutrition status conditions [20]. Recent research has demonstrated that IoT-based BMI prediction models, equipped with machine learning algorithms, provide highly accurate and efficient assessments compared to traditional manual methods [21]. These advancements leverage sensor technology and cloud-based AI systems, enabling real-time analysis and reducing errors in nutritional assessments. Furthermore, AI-driven applications in healthcare and nutrition have expanded, allowing wearable and mobile devices to monitor health metrics and predict potential risks using machine learning models [22]. The growing adoption of IoT and AI in nutritional healthcare highlights the potential for personalized dietary interventions and early disease prevention strategies, ultimately improving public health outcomes.

The integration of XAI in healthcare has enhanced nutritional assessment and prediction, with an XAI-based clinical decision support achieving high accuracy in diagnosing malnutrition and related geriatric syndromes [23]. Similarly, deep neural networks using SHAP-based interpretability have improved blood glucose prediction for Type 1 diabetes mellitus by incorporating meal-related nutritional factors [24]. Transformer-based models have further demonstrated their superiority in analyzing large-scale nutritional datasets, identifying key dietary factors influencing Alzheimer's disease mortality, outperforming traditional machine learning approaches [25]. These advancements highlight the growing role of XAI in precision nutrition, enabling personalized dietary recommendations and improved clinical decision-making.

The integration of ML models in nutritional assessments has demonstrated promising results in recent studies. A study by Ref. [26] employed Naïve Bayes and logistic regression to predict vitamin D deficiency using anthropometric parameters such as waist circumference (WC), body mass index (BMI), and body roundness index (BRI). While the models identified sex-specific predictors and offered moderate predictive accuracy, the choice of only Naïve Bayes and logistic regression restricts a broader comparison with more advanced model. Similarly, another work [27] used the same models on a cohort, emphasizing their utility in nutritional epidemiology but highlighting the limitations of not incorporating more robust algorithms like deep learning networks. These studies underscore the need for more comprehensive datasets and advanced techniques to improve prediction accuracy and generalizability. In the context of child malnutrition, studies have leveraged a broader range of ML methods. A comparison of Random Forest, Gradient Boosting, Support Vector Machine (SVM), and logistic regression for predicting child malnutrition outcomes achieved high accuracy, with Random Forest and Gradient Boosting exceeding 97 % [6]. However, these methods often lacked scalability and real-world applicability due to overfitting risks and computational demands. Another study [28] compared C4.5 Decision Tree, K-Nearest Neighbors (KNN), and Naïve Bayes for predicting nutritional status in children under five. Although the C4.5 Decision Tree performed best (89.87 % accuracy), the small dataset, only 360 child patients, and limited features (age, gender, height, weight) restricted its broader applicability. To improve malnutrition prediction in sub-Saharan Africa, a hybrid ensemble approach combining decision trees, SVM, and logistic regression achieved 94 % accuracy by leveraging the strengths of each method [29]. Despite its strong performance, the complexity of the hybrid model and lack of interpretability presented challenges for deployment in low-resource settings. Similarly, a study targeting Ethiopian zones [30] identified key determinants of undernutrition using Random Forest and Gradient Boosting, achieving an accuracy of 89 %. However, the study did not explore real-time monitoring technologies, such as IoT, which could enhance early detection and scalability. The study on classifying and predicting children's nutritional status in Ethiopia using Long Short-Term Memory-Fully Connected (LSTM-FC) has been achieved 93 % of accuracy. It emphasizes rigorous data preprocessing, including handling missing values, normalization, and balancing, to enhance model performance [31]. The study using Method for Extremely Rapid Observation of Nutritional Status (MERON), a CNN-based AI tool for mal-nutrition status prediction [32]. The model achieved a 78 % accuracy in predicting adult BMI classifications and successfully classified 60 % of cases in a sample of 3167 Kenyan children.

Although these studies demonstrate the potential of ML in nutritional assessments, but they face some challenges. Traditional models, such as Naïve Bayes, logistic regression, and decision trees, are limited in their capacity to uncover complex patterns and dependencies within high-dimensional dataset, the challenges that CNN and LSTM models are specifically designed to address. However previous studies using LSTM and CNN in Refs. [31,32] fail to fully leverage complex cross-feature dependencies, limiting their effectiveness in modeling intricate relationships between anthropometric and demographic indicators. While ensemble methods like Random Forest and Gradient Boosting offer high accuracy, their limited interpretability prevent their practical deployment in real-world scenarios. To overcome these limitations, this study integrates CNN and LSTM models with MHA and XAI, providing a robust framework that enhances predictive accuracy, interpretability, and scalability for nutritional status prediction. The incorporation of XAI further enhances transparency by providing feature level insight, allowing health-care professionals to understand and validate the model's decision-making process. Notably, the XAI results validate the model's consistency with domain knowledge, emphasizing the importance of z-scores aligned with WHO standards for nutritional assessment [33,34] as critical features. This alignment with established health metrics enhances the model's reliability and real-world applicability.

3. Material and method

This study presents a deep learning-based method for predicting children's nutritional status, integrating MHA and XAI to enhance both accuracy and interpretability. As depicted in Fig. 1, the proposed framework incorporates extensive data preprocessing, hyperparameter optimization, and MHA, ensuring a robust and transparent approach to nutritional assessment. The dataset collected from primary sources, includes key anthropometric features such as height, weight, upper arm circumference (UAC), head circumference (HC), along with contextual factor, such as parenting practice (e.g., parents, grandparents, day care and others), and age.

The methodology consists of several key stages, including data preprocessing (feature selection, handling missing values, feature scaling, and correlation removal for redundant features), model development using CNN and LSTM architectures, hyperparameter tuning via Grid Search, and attention-based feature prioritization using MHA. The final model is rigorously evaluated, and its interpretability is assessed through XAI technique, particularly SHAP, ensuring that predictions align with WHO-standard Z-scores for nutritional assessment. This XAI-driven approach provides transparent, clinically interpretable insights, facillitating more actionable decision-making in the domain of child health.

3.1. Input

Fig. 2 ilustrates the IoT-based data acquisition framework for collecting anthropometric and demographic data. The anthropometric measurement process is tailored for three categories: (a) Infants, requiring specialized non-contact or minimal-contact measurement devices (b) Toddlers and children unable to stand, utilizing assisted anthropometric tools, and (c) Toddlers and children able to stand. The sensor-based platform consists of Rotary Encoder Sensors and Load Cell Sensors, capturing precise weight, height, and body proportions. A Microcontroller Processing Unit processes this data and transmits it to cloud storage via HTTP/MQTT protocols, enabling secure real-time data fusion. Additionally, demographic data (e.g., age, gender, parenting practice) is retrieved through API-based integration from the health department's database. The system automatically synchronizes sensor measurements with demographic attributes in the cloud, where the data is preprocessed, stored, and merged into a unified format before being used for deep learning-based analysis. This cloud-based synchronization ensures scalability, accessibility, and efficient handling of large datasets for AI-driven nutritional predictions.

3.2. Extensive data preprocessing

While data preprocessing and feature selection are standard in AI pipelines, our approach uniquely tailors these steps to structured anthropometric data, optimizing input reshaping for attention mechanisms and ensuring that retained features align with both model performance and clinical relevance, as validated through SHAP.

a Feature Selection and Engineering

Feature selection is a critical step in ML that enhances model performance by identifying the most influential features while eliminating irrelevant or redundant ones to reduce complexity and improve generalizability [35]. This study leverages WHO standard z-scores, including ZS_WA (weight for age z-score), ZS_HA (height for age z-score), and

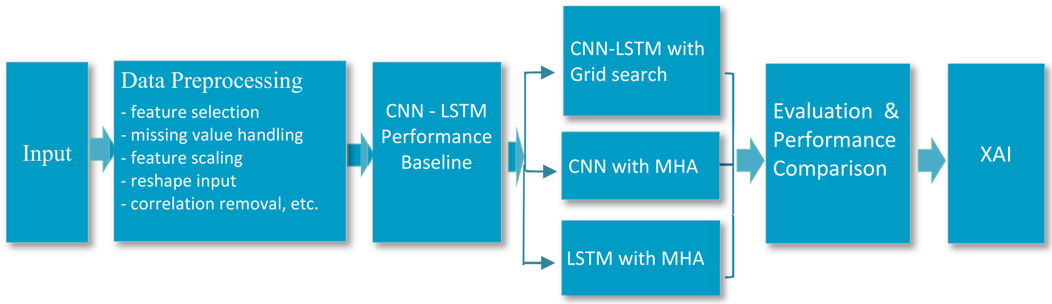

Fig. 1. The proposed framework integrating preprocessing, attention-based models, and XAI for nutritional status prediction.

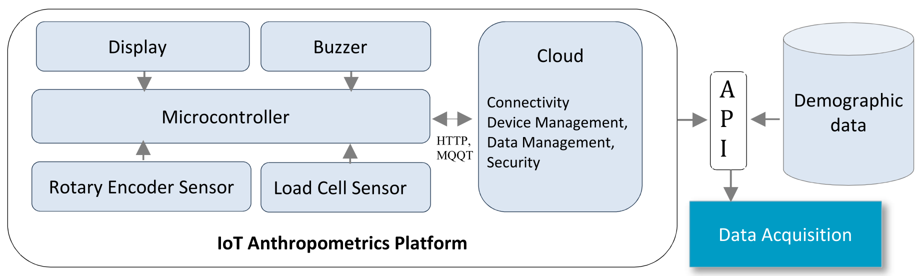

Fig. 2. The architecture of data acquition integrating IoT anthtropometric measurements and demographic data.

ZS_WH (weight for height z-score), which are globally recognized as benchmark for child nutritional assessment. These z-scores are derived from anthropometric measurements ensuring cross-population comparability and model generalizability. The features along with UAC, parenting practice and related features were carefully examined. They were evaluated using a combination of statistical methods and ML techniques to ensure their relevance and impact on the prediction task. To minimize redundancy, features with high inter-correlations were systematically removed, as highly correlated features can introduce multicollinearity [36], which may adversely affect model performance.

To further refine feature selection, we applied Random Forest Feature Importance (RFI) to quantify the predictive contribution of each feature. RFI is calculated based on the decrease of impurity using Mean Decrease Impurity (MDI) criterion [37], which measeure the reduction in Gini impurity or entropy when a feature is used for splitting in a decision tree. The feature importance I_j of a feature j is calculated as follows:

$$I_j = \frac{1}{T}\sum_{t=1}^{T}\sum_{s\in S_t^j}\Delta I_s \tag{1}$$

Where:

- T is the total number of trees in the Random Forest.
- S_t^j is the set of all nodes s in tree t where feature j is used for splitting.
- ΔI_s is the decrease in impurity Gini impurity or entropy at node s.

The decrease in impurity at a specific node s is given by:

$$\Delta I_s = I_{\text{parent}} - \left(\frac{N_{\text{left}}}{N_{\text{parent}}} I_{\text{left}} + \frac{N_{\text{right}}}{N_{\text{parent}}} I_{\text{right}}\right) \tag{2}$$

Where:

- I_{parent} is the impurity of the parent node.
- $I_{\text{left}}, I_{\text{right}}$ are the impurities of the left and right child nodes.
- $N_{\text{parent}}, N_{\text{left}}, N_{\text{right}}$ are the number of samples in the parent, left, and right nodes, respectively.

While RFI identifies globally important features, it does not capture dynamic feature interactions that vary by individual nutritional status in the dataset. To address this limitation, we incorporate MHA within CNN and LSTM models. Unlike static feature selection, MHA dynamically learns feature dependencies, ensuring that critical anthropometric relationships are prioritized during training. This enhances prediction accuracy, particularly for complex nutritional patterns that evolve over time. Furthermore, SHAP were applied post-training to validate feature importance rankings and ensure alignment with WHO-standard nutritional indicators. By integrating both pre-training (RFI) and post-training (SHAP) feature validation methods, our study ensures that the selected features contribute both statistically and clinically to the prediction of child nutritional status.

b Missing Value Handling

Handling missing values is a critical preprocessing step to ensure the robustness and reliability of ML models [38], as incomplete data can lead to biased results or errors during model training. In this study, a systematic approach was applied to address missing values based on the nature of the data:

(1) Numerical features (e.g., weight, height, Z-scores) were imputed using the column mean. This approach preserves the distribution of the data, prevents skewness and ensure dataset integrity for analysis [39].

(2)Categorical features such as parenting practice were imputed using mode substitution, minimizing the distortion of category frequencies and ensuring alignment with the overall data distribution [40]. Given that critical anthropometric features influence nutritional status predictions. Proper imputation strategies were carefully applied to minimize data loss and ensure the dataset remained suitable for model training. This careful preprocessing step plays a vital role in maintaining the integrity of the dataset, particularly in health-related studies where data completeness directly impacts outcome reliability.

c Feature Scaling

Feature scaling ensures that all numeric features are on a similar scale, which is particularly important for deep learning models. In our case, features such height (cm), weight (kg), and z-scores had different ranges and units, which could introduce bias the models trainning. To address this, We applied standardization (mean = 0, standard deviation = 1) to normalize all numerical features, ensuring optimal model performance. Standardization is a critical preprocessing step, particularly for models that rely on magnitude-based relationship [41], such as CNN, LSTM and distance-based algorithm. Unlike Min-Max normalization, which rescales data within a fixed range (e.g., [0,1]), standardization was chosen because it is more robust to outliers and ensures that features with naturally higher magnitudes (e.g., weight vs. Z-scores) do not dominate the learning process. This is particularly crucial for CNN (spatial) and LSTM (sequential), which are sensitive to variations in feature scaling. Additionally, feature scaling plays a key role in MHA mechanism, which dynamically assigns importance to different features. Without proper standardization, features with larger values (e.g., weight) could bias attention weights, leading to distorted feature prioritization. By ensuring a uniform scale, MHA can learn unbiased feature relationships, improving predictive accuracy.

d Reshape Input

Reshaping for CNN and LSTM prepares the dataset into the required input dimensions for these architectures [9]. For CNNs, the input data, which is initally a 2D array of features, e.g., *[n_samples, n_features]* was reshaped into a 3D structure by adding a channel dimension, e.g., *[n_samples, n_features, 1]*. This allows CNNs to learn spatial relationships between features, such as patterns among z-scores or correlations between anthropometric indicators and contextual variables. For LSTMs, the dataset was transformed into a 3D format, e.g., [n_samples, timesteps, n_features]. Although the dataset does not contain inherent temporal information, features were structured as a sequence, allowing LSTM to capture feature dependencies dynamically [42]. For example, z-scores, UAC, one-hot encoded categories were reshaped into a "time-step-like" format such as [1,10,1]. This structure forces LSTM to interpret feature dependencies sequentially, which is especially useful if feature order influences nutritional assessment (e.g., Z-scores followed by UAC and parenting practices). LSTM uses the gated mechanism to prioritize and learn from certain feature relationships [43]. Furthermore, reshaping is essential for MHA, which dynamically learns feature interactions. Proper input structuring enables MHA to effectively identify cross-feature relationships, ensuring that important anthropometric indicators are learned adaptively.

e Correlation Removal

We conducted a thorough analysis to identify highly correlated features, which can lead to multicollinearity and degrade model performance [44]. A correlation heatmap was used to detect pairs of features with a correlation coefficient exceeding 0.85, a commonly used threshold in ML studies. High correlations indicate that certain features are providing overlapping or redundant information, which can inflate model complexity and reduce generalization capability [45,46]. By removing highly correlated features, we ensured that CNN and LSTM models learned independent feature relationships, preventing bias from redundant indicators. This step was particularly crucial for MHA, as removing redundant features allows MHA to assign attention weights more selectively. Instead of distributing attention to multiple correlated features, MHA focuses on unique and informative features, leading to better feature interpretability and a more compact representation.

f Categorical Encoding

Non-numeric features, such as parenting practice, were preprocessed to ensure compatibility with ML models by transforming them into numerical representations. One-Hot Encoding technique was chosen over Label Encoding due to its nominal nature, ensuring that each category was treated independently and equitably during model training [47]. One-Hot Encoding was selected to prevent CNN and LSTM from misinterpreting categorical variables as ordinal data, a common issue when using Label Encoding. By assigning each category a separate binary column, one-hot encoding ensures that the model does not assume hierarchical relationships between different parenting practices, preserving data integrity. Proper categorical encoding is crucial for MHA, as it ensures that each category is treated as an independent feature, preventing one category from dominating attention weight distribution. This allows MHA to learn unbiased feature relationships, improving interpretability and robustness in nutritional predictions.

g Outlier Handling

In this study, we carefully addressed outliers in key anthropometric features, including height, weight, and UAC, using the Interquartile Range (IQR) method [48]. Outlier handling is particularly crucial for MHA, as extreme values can distort attention weight distributions, leading to biased feature importance scores. By removing statistical anomalies, MHA can focus on meaningful cross-feature interactions, improving the interpretability of anthropometric indicators. Therefore, removing outliers enhances CNN and LSTM robustness by preventing extreme values from causing unstable weight updates during backpropagation. This ensures faster model convergence and better generalization to unseen data.

h Class Imbalance Handling

The dataset used in this study exhibited a significant class imbalance, where certain classes, such as *normal nutrition*, had substantially more samples compared to minority classes like *overnutrition* and *undernutrition*. To address the imbalance, Synthetic Minority Over-sampling Technique (SMOTE) was applied to oversample the minority classes [49]. By balancing class distributions, SMOTE ensures that MHA assigns attention weights fairly across all nutritional categories rather than being biased toward dominant classes. This improves model interpretability and feature importance consistency. By applying SMOTE, the model learns more representative patterns, ensuring higher classification accuracy across all nutritional risk levels.

3.3. Algorithms

a. CNN were employed to extract meaningful features from the dataset [13], which was strategically transformed into a suitable representation for deep learning. The dataset comprises key anthropometric features, such as height, weight, UAC, HC, and z-scores like 'z-score weight for age' (ZS_WA), 'z-score height for age (ZS_HA), 'z-score weight for height' (ZS_WH), alongside parenting practice as a contextual factor. Unlike traditional fully connected network, which treat each feature independently, 1D Convolutions (Conv1D) efficiently learn spatial-like feature relationships in structured tabular data. This is particularly important in nutritional assessments, where anthropometric measurements interact in complex ways. By applying Conv1D, CNN captures local dependencies and critical patterns [50], enabling the model to uncover non-linier relationships between anthropometric indicators and children's nutritional status. In the preprocessing stage, highly correlated features were removed to reduce redundancy and avoid multicollinearity [45]. This ensures that only the most relevant features are fed into the model, improving efficiency. After pre-processing, the 1D convolutional layers of the CNN extract feature dependencies dynamically. For example, CNN learns how ZS_HA, ZS_WA and UAC jointly influence nutritional risk level. While CNN efficiently captures feature relationships, it does not inherently distinguish the most critical dependencies. To address this, MHA was integrated, enabling CNN to prioritize key anthropometric interactions dynamically rather than treating all local dependencies equally.

b. LSTM networks were employed to capture sequential dependencies within the dataset [9,13], which includes key anthropometric measurements such as height, weight, UAC, HC, and z-scores, alongside behavioural factor like parenting practice. Unlike CNN which focus on spatial relationship between features, LSTM is specialized for modeling dependencies over sequential-like data. It makes LSTM highly effective for structured tabular data where relationships between features can be treated as a sequence [13,42]. Although the dataset does not explicitly contain time-series information, anthropometric features follow logical progression trends (e.g., weight-for-age vs. height-for-age), allowing LSTM to learn dynamic patterns relevant to nutritional status prediction. LSTM was particularly effective in analyzing interdependencies between anthropometric and contextual features. Sequential relationships between height, weight, and Z-scores were effectively captured, helping the model identify early signs of malnutrition or abnormal growth trajectories. Furthermore, LSTM modeled feature contributions over multiple steps, improving the detection of undernutrition risk factors such as UAC and HC measurements. However, LSTM applies equal importance to all time steps, which can limit its ability to prioritize the most critical feature dependencies. To enhance LSTM's attention to key anthropometric indicators, MHA was integrated, enabling the model to dynamically assign varying attention weights to different feature dependencies. This ensures that the model learns to focus on the most influential sequential patterns, such as Z-score variations and weight-for-age progression.

c Hyperparameter Optimization

Grid Search is an effective method to optimize the hyperparameters of CNN and LSTM models [51]. The performance of deep learning models depends heavily on hyperparameters such as the batch size which affects training speed and convergence, the Learning rate that controls the step size during weight updates, number of epochs that defines the number of times the model iterates over the entire dataset, and related parameters. By using Grid Search, the algorithm systematically test combinations of these hyperparameters to find the optimal configuration that yields the highest accuracy [51]. This approach provides a systematic comparison of model performance, ensuring the best results for further step in predicting child nutritional status. Fig. 3 depicts the workflow of the grid search.

For small dataset, grid search may lead to overfitting by selecting hyperparameters that are too specific to the training set but fail on unseen data [52]. It also requires significant computational time, especially for complex models like CNN and LSTM, as the number of hyperparameter combinations grows exponentially, e.g., testing 4 batch sizes, 3 learning rates, and 20 epochs results in 240 models. Hence, for a lower computational cost, we also propose an attention mechanism approach [53].

d CNN with MHA

Hyperparameter tuning via Grid Search Optimization is essential for fine-tuning hyperparameters. It systematically evaluates combinations of hyperparameters to find the configuration that yields the best overall performance for a model. However, hyperparameter tuning alone does not address the model's ability to focus on the most relevant features within the dataset [54]. This limitation becomes particularly significant for datasets where certain features contribute disproportionately to the outcome [43], as may occur in nutritional status dataset. To address this, MHA is integrated into both CNN and LSTM architectures, enhancing feature prioritization by dynamically assigning importance weights to relevant inputs [55]. CNN is highly effective at extracting feature representations, making them well-suited for learning complex relationships in structured datasets. However, standard CNN architectures rely on fixed-size convolutional filters, which primarily capture local dependencies, limiting their ability to model global feature interactions in tabular data like nutritional status indicators. Given that nutritional assessment depends on interactions between non-adjacent anthropometric features such as height, weight, UAC, and Z-scores, traditional CNNs may fail to emphasize critical feature relationships. To address this limitation, MHA is integrated into CNN, allowing the model to dynamically assign importance weights to different features, improving both accuracy and interpretability. This enhancement ensures that CNN effectively learns cross-feature dependencies. Given a set of input features example, e.g., Heigh, Weight, UAC, HC, and Parenting_practice (Parenting_P),

Fig. 4 depicts the block chart for CNN with MHA. It clearly illustrates how MHA is integrated into CNN after Global Average Pooling (GAP), improving the model's ability to focus on key nutritional indicators.

The MHA Block in CNN consists of:

1 Query, Key, and Value Transformation: The output of GAP from the Conv1D layers serves as the input to MHA. Each feature (e.g., weight, height, UAC, HC, …) is projected into different subspaces through linear transformations to generate queries (Q), keys (K), and values (V).
2. Scaled Dot-Product Attention: This mechanism ensures that features with stronger correlations receive higher attention scores. Each attention head computes attention weights using:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

where:

- o Q (Query), K (Key), V (Value): matrices represents of inputs.
- o QK^T: Computes the similarity between the query (Q) and key (K) matrices.
- o $\sqrt{d_k}$: Scaling factor, where d_k is the dimensionality of the key vectors.
- o softmax$(\cdot)$: Normalizes the scores to ensure they sum up to 1, transforming them into attention weights that determine how much influence each feature has on the final output.
- o Multiplying by V: The final weighted sum is computed, where the most relevant features (those with the highest attention scores) are emphasized.

3. Multiple Attention Heads: multiple parallel heads to learn different aspects of feature relationships. Each head captures unique feature dependencies (e.g., one head focus on height-weight correlation, another on HC and UAC).

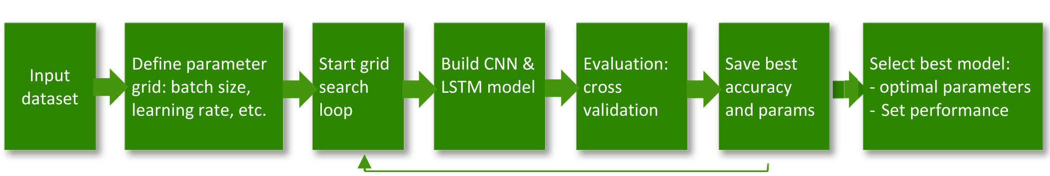

Fig. 3. Grid search optimization block chart.

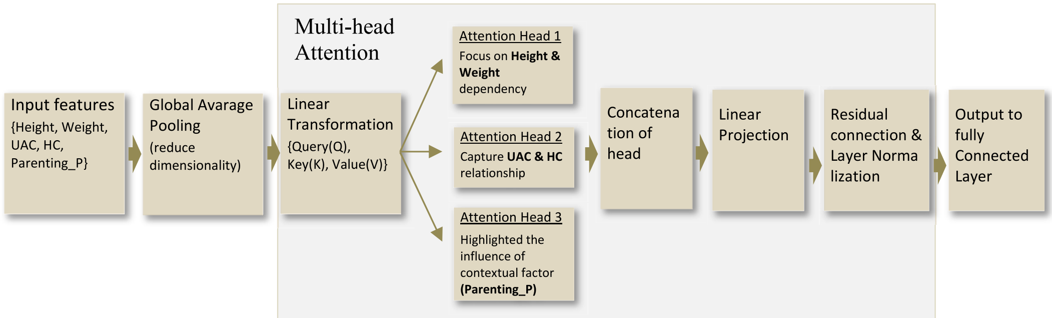

Fig. 4. Block chart of MHA in CNN.

4. Concatenation & Linear Transformation: the outputs from all heads are concatenated and passed through a linear transformation layer to integrate the learned feature representations.
5. Residual Connection & Layer Normalization: To stabilize training, a residual connection adds the original input before applying layer normalization, ensuring smooth gradient flow.

Therefore, CNN with MHA is employed to enhance the contribution of specific feature channels that play a pivotal role in classification [56]. Channel attention works by computing global statistics, e.g., average pooling across the dataset dimensions, and applying a learnable weighting mechanism [55,57].

e LSTM with MHA

For LSTM, AHM is applied to focus on the most relevant features from the sequential outputs of the LSTM layer [55]. Although the dataset is tabular, the LSTM treats the features as pseudo-timesteps, where each timestep represents one feature dimension [13]. AHM assigns weights to these pseudo-timesteps through a dense layer with a softmax activation, ensuring that the most critical features, for instance, height, weight, and parenting practice receive higher attention during decision-making. This allows LSTM to emphasize the features that contribute most significantly to the classification of nutritional status. The integration of the Attention mechanism into LSTM enhances the models' ability to adaptively learn feature relevance [55], particularly in datasets where the interplay of anthropometric indicators and health features determines nutritional outcomes. While both CNN with MHA and LSTM with MHA employ attention mechanisms, they serve distinct roles in processing structured datasets. CNN with MHA captures spatial dependencies using 1D convolutions and refines feature importance through attention after Global Average Pooling, operating on a 3D tensor representation (samples, features, 1) to prioritize local feature interactions. LSTM with MHA, in contrast, processes data in a (timesteps, features) format, assigning dynamic attention to cross-feature relationships rather than fixed spatial structures. The comparison of CNN with AHM and LSTM with AHM, particularly the problems addressed in this study presented in Table 1.

3.4. *Evaluation & visualization*

The evaluation and visualization stage is a critical component of the workflow, as depicted in Fig. 1, where the performance of CNN and LSTM models, both with and without optimized hyperparameters, and also with MHA are rigorously assessed. This process ensures the reliability of the models in predicting children's nutritional status based on the dataset, which includes features such as anthropometric measurements and contextual factors, i.e., parenting practice.

a. Evaluation Metrics

Table 1
The differences of MHA aspects integrated with CNN and LSTM.

Aspect	CNN with MHA	LSTM with MHA
Feature Interpretation	Operates on spatial feature relationships, emphasizing local dependencies in features	Works on pseudo-sequential dependencies, refining how features contribute over different steps.
Embedding and Feature Encoding	Directly processes feature maps from Conv1D layers, without requiring embedding layers.	Requires embedding layers to structure the input before passing through attention.
Attention Application	Uses channel attention to prioritize important feature maps for classification.	Applies sequential attention, allowing dynamic feature weighting across time steps.
Output Processing	Passes attention-enhanced features to dense layers for final classification.	Feeds attention-enhanced sequences into LSTM layers for long-term feature learning.
Problem addressed	Addresses the need for feature prioritization in spatial feature extraction, improving CNN's performance.	Ensures that LSTM prioritizes critical features in nutritional predictions

Given the dataset and the algorithms, the standard accuracy metrics alone is insufficient. To provide a comprehensive evaluation, the following metrics are used [58]: (1) accuracy: A basic measure of the model's overall prediction correctness but considered in conjunction with other metrics due to class imbalance; (2) Precision, Recall, and F1-Score: Precision ensures the model minimizes false positives, particularly for high-risk classes. Recall emphasizes the ability to identify all true positive cases, ensuring vulnerable children are not overlooked. F1-score balances precision and recall, providing a robust measure for imbalanced datasets.

b. Visualization

To facilitate insights into the model's behaviour, the following visualizations are provided: (1) Metric performance comparisons, i.e., line graphs comparing accuracy, precision, recall, and F1-scores across models highlighting improvements achieved with the three scenarios. (2) SHAP summary plot for each class, namely, Normal, Undernutrition and Overnutrition.

3.5. *Explainable AI using SHAP*

SHAP was integrated into the workflow as the final stage to provide interpretability for the deep learning models [59]. SHAP, based on a game-theoretic approach, explains how each input feature contributes to model predictions, making deep learning models more explainable in decision-making processes [17]. This is particularly crucial in child nutritional assessment, where understanding how anthropometric and contextual factors influence predictions fosters trust and informed intervention strategies.

- Feature Importance: SHAP values quantify the influence of each input feature on the model's predictions [60]. For example, SHAP can highlight how z-scores or parenting practice affects the prediction of categories like undernutrition.
- Global Interpretability [61]: Aggregated SHAP values provide a global view of feature importance across the dataset, identifying key factors driving predictions for all samples.
- Local Interpretability [62]: For each class prediction, SHAP explains why a particular sample was classified into a specific class. For instance, it may reveal that high values of ZS_HA (Height-for-Age Z-scores) and UAC were the primary contributors to the prediction of Undernutrition class.

For health practitioners, they use SHAP visualizations to understand which features have the most influence on specific predictions, supporting targeted interventions for Undernutrition children class. The transparency provided by SHAP fosters trust in the predictive model, which is critical for adoption in real-world healthcare settings [63,64]. Furthermore, SHAP values can identify biases or issues in the model by revealing unexpected feature contributions, enabling refinement and optimization [65]. By integrating SHAP with MHA enhanced deep learning architectures, this study ensures that feature prioritization is not only driven by learned attention weights but is also validated post hoc using explainability metrics. This alignment with domain knowledge strengthens the reliability of the model in child health monitoring and decision-making. Therefore, combining robust metrics, insightful visualizations, and feature-level explanations, this approach ensures the models are both accurate and interpretable, making them highly valuable for health monitoring and decision-making [66].

Fig. 5 illustrates the integration of SHAP with the nutritional status prediction model to enhance interpretability and feature importance analysis. The workflow begins by wrapping the attention-enhanced model with SHAP, ensuring compatibility while maintaining feature structure. The SHAP Explainer computes feature contributions using a summarized dataset, quantifying the impact of anthropometric and contextual variables on predictions. The final visualization phase generates global explanations, identifying key predictors, and local explanations, clarifying individual predictions for healthcare professionals. This approach enhances model transparency, supporting data-driven decision-making in child nutrition assessment. By integrating XAI into the optimized predictive model [61], this workflow bridges the gap between model accuracy and interpretability, empowering healthcare professionals to harness the full potential of AI in public health initiatives.

4. Experimental evaluation

All experiments were conducted using Python 3.8.19. Deep learning models were implemented using TensorFlow v2.13 and the Keras v2.13.1. Preprocessing and performance evaluation utilized the scikit-learn library v1.3, while SHAP v0.44.1 was employed for feature attribution and explainability analysis.

4.1. Dataset description and hyperparameter setup

The dataset consists of 9605 samples collected from primary sources, integrating anthropometric and contextual attributes relevant for predicting children's nutritional status. The dataset attributes are categorized as follows (see Appendix Table 4 for details):

1. Anthropometric Measurements: Weight, Height, Upper Arm Circumference (UAC), Head Circumference (HC).
2. Z-Score Indicators (WHO Standards): Weight-for-Age Z-score (ZS_WA), Height-for-Age Z-score (ZS_HA), and Weight-for-Height Z-score (ZS_WH).
3. Derived Ratios and Contextual Features: UAC_to_Height (UAH), Weight_to_Age (WA), Age, Parenting Practice.
4. Nutritional Status Labels (One-Hot Encoded): Normal, Under Nutrition, Overnutrition.

The dataset was split into 70 % of training, 30 % of test sets. To ensure the optimal performance, Grid search was employed to fine-tune the hyperparameters of each algorithm. Table 2 presents set of selected optimizable hyperparameters for LSTM and CNN using Grid Search algorithm.

Batch size, which determines the number of samples for weight updates in each iteration, was kept consistent across both models to ensure uniform training dynamics. Learning rate was carefully tuned, with lower values, e.g., 0.001 providing stability for LSTM's sequential pattern learning and higher values, e.g., 0.01 accelerating CNN convergence due to its efficient feature extraction [55].

In LSTM, the number of units in the hidden layers was optimized to capture complex temporal dependencies between features like height, weight, and z-scores, while in CNN, the number of convolutional filters allowed the model to extract relationships among feature. Therefore, these carefully tuned hyperparameters not only enhanced model accuracy but also improved generalization.

4.2. Algorithm performance

Grid search provides a systematic approach to explore the hyperparameter space, enabling the identification of the best combinations for optimal performance. Unlike grid search, which optimizes parameter settings, attention enhances the model's capability to focus on multiple key patterns simultaneously, improving interpretability in complex datasets [53]. Table 3 provides a detailed performance evaluation of CNN and LSTM models under three scenarios: (1) default parameters with extensive data-preprocessing, (2) hyperparameter optimization through grid search, and (3) CNN and LSTM with AHM.

Table 2
CNN and LSTM hyperparameters selected for optimizati on.

Hyperparameter	LSTM	CNN
Batch Size	[16, 32]	[16, 32]
Learning Rate	[0.001, 0.01]	[0.001, 0.01]
Dropout Rate	[0.2, 0.5]	[0.2, 0.5]
Number of Units/Filters	[32, 64]	[16, 32]

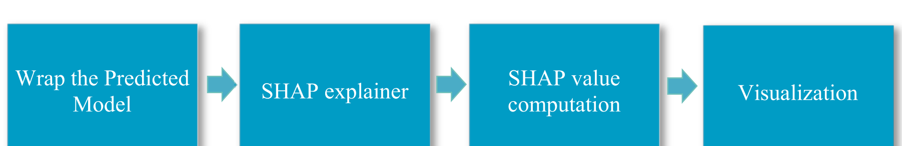

Fig. 5. The workflow of SHAP integration with model prediction.

Table 3
Model performance with the three scenarios (%).

	Extensive data preprocessing		Grid Search Parameters Optimization				AHM			
	CNN	LSTM	CNN	± %	LSTM	± %	CNN	± %	LSTM	± %
Accuracy	97,74	96,26	98,46	0,74	98,93	2,77	99,08	1,37	98,91 %	2,75
Precision	97,81	96,31	98,10	0,30	98,56	2,34	99,09	1,31	98,92 %	2,71
Recall	97,74	96,26	90,46	−7,45	93,13	−3,25	99,08	1,37	98,91 %	2,75
F1-Score	97,74	96,26	93,88	−3,95	95,64	−0,64	99,08	1,37	98,91 %	2,75

4.3. Enhancement of accuracy through MHA

Fig. 6 illustrate the enhancement of performance accuracy through MHA. CNN with MHA achieves an accuracy of 99.08 %, outperforming both baseline CNN (97.74 %) and Grid Search CNN (98.46 %), indicating that spatial feature learning combined with attention mechanisms allows the model to better capture critical anthropometric relationships. While LSTM also benefits from MHA, achieving an accuracy of 98.91 %, its performance gain is less pronounced compared to CNN, suggesting that attention-enhanced spatial feature extraction is more effective for tabular anthropometric data than sequential dependency modeling. These findings demonstrate that the proposed adaptation of MHA for structured healthcare data introduces a methodological enhancement that improves predictive accuracy beyond standard CNN and LSTM approaches. The improvements observed in CNN with MHA align with prior research indicating that attention mechanisms are particularly beneficial in scenarios where feature importance varies dynamically across input data, reinforcing the model's ability to learn representations of nutritional status. To further validate the effectiveness of MHA, we incorporate SHAP-based explainability analysis, ensuring that the model's enhanced accuracy aligns with established nutritional indicators, thereby addressing concerns related to interpretability and model transparency.

4.4. Comparative analysis of CNN and LSTM with MHA

Fig. 7 depicts the comparative analysis of CNN and LSTM with MHA. It reveals distinct performance trends across accuracy, precision, recall, and F1-score, highlighting the impact of spatial versus sequential feature learning.

In terms of accuracy, CNN with MHA (99.08 %) consistently outperforms LSTM with MHA (98.91 %), indicating that CNN-based attention mechanisms more effectively capture the spatial dependencies within anthropometric data. Similarly, CNN with MHA achieves a precision score of 99.09 %, exceeding LSTM with MHA's 98.92 %, reinforcing that spatial feature prioritization through attention enhances classification confidence. However, recall exhibits a notable divergence, where LSTM with MHA (98.91 %) slightly surpasses CNN with MHA (99.08 %), suggesting that LSTM benefits more from attention when capturing long-term dependencies in sequential health indicators. The F1-score for both models follows a similar pattern, with CNN with MHA reaching 99.08 % and LSTM with MHA attaining 98.91 %, emphasizing that CNN's ability to preserve both precision and recall leads to a more balanced performance.

Compared to baseline models, CNN and LSTM without MHA exhibit lower scores across all metrics, validating that MHA significantly enhances deep learning-based nutritional status prediction. While CNN-MHA achieves the highest overall accuracy and precision, LSTM-MHA remains competitive, particularly in recall, indicating its advantage in identifying sequential dependencies. These findings provide empirical evidence supporting the superiority of attention-enhanced CNN for structured anthropometric data, while also demonstrating that LSTM with MHA remains beneficial for capturing temporal aspects of nutritional health patterns.

To provide further insight into the model's predictive performance, Fig. 8 presents examples of input data and their corresponding predictions made by the proposed CNN model with MHA. These examples showcase the first five feature values of selected inputs after preprocessing. The one-hot encoded actual label [1 0 0] corresponds to 'Normal', [0 1 0] corresponds to 'Overnutrition', and [0 0 1] corresponds to 'Undernutrition'. For Sample 1, the predicted label is 'Overnutrition' with a confidence vector of [1.2566075e-09, 9.9999988e-01, 1.5408440e-07], corresponding to the label representation [0 1 0]. For Sample 2, the predicted label is 'Normal' with a confidence vector of [9.999999e-01, 8.436247e-08, 2.870198e-10], corresponding to the label representation [1 0 0]. In both samples, the predicted label matches the actual class, with the confidence values demonstrating extremely high certainty for the correct class in each case.

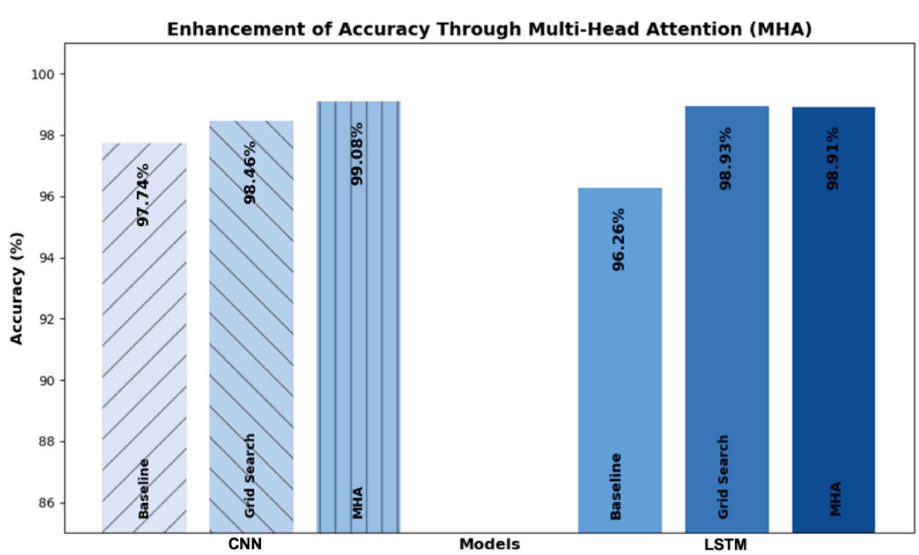

Fig. 6. Accuracy improvement of CNN and LSTM with MHA.

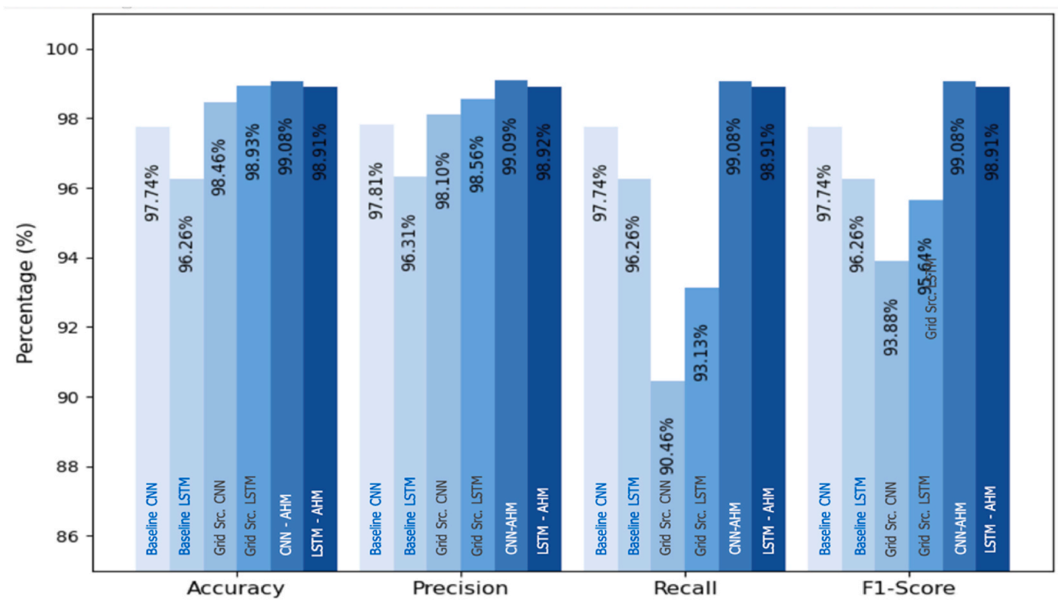

Fig. 7. Comparative performance of CNN and LSTM with MHA.

```
Sample 1:
Input Features:
  Weight: 0.001242
  Height: 0.675317
  UAC: 0.031931
  HC: 0.903922
  ZS_WA: 0.632388
Actual Class: 1 ([0 1 0])
Predicted Class: 1 ([1.2566075e-09 9.9999988e-01 1.5408440e-07])
```

```
Sample 2:
Input Features:
  Weight: 0.001286
  Height: 0.800000
  UAC: 0.028183
  HC: 0.903922
  ZS_WA: -0.740000
Actual Class: 0 ([1 0 0])
Predicted Class: 0 ([9.999999e-01 8.436247e-08 2.870198e-10])
```

Fig. 8. Samples of actual class output in response to some input features.

For visualization using SHAP, CNN with MHA is more suitable as it captures spatial relationships, making SHAP's feature importance and dependence plots easier to interpret. SHAP can clearly highlight how individual features, such as height or weight, contribute to the predictions, leveraging CNN's structured representation of input data.

4.5. XAI using SHAP

Based on the results, the CNN with MHA emerged as the superior algorithm, particularly when accuracy and F1-score were the key metrics of evaluation. Consequently, CNN with MHA was selected as the preferred algorithm for integrating XAI techniques using SHAP. This decision is justified by CNN's robust predictive performance and its ability to capture complex patterns in the data, such as interactions between anthropometric features and contextual factors. The integration of SHAP with the attention-augmented CNN model aims to provide interpretable insights into the decision-making process, clearly highlighting the contribution of each feature to the predictions.

The SHAP summary plot for the class 'Normal' as illustrated in Fig. 9 provides insights into how individual features influence the model's predictions. For feature importance, The features are ranked by their overall impact on the model's output. From the figure, **ZS_WH** and **Height** have the most significant impact, suggesting they are the primary contributors to the prediction of the 'Normal' class. Other key features include **ZS_WA** and **ZS_HA**, indicating the relevance of anthropometric indices in determining nutritional status. SHAP values on the x-axis indicate the direction and magnitude of a feature's impact,

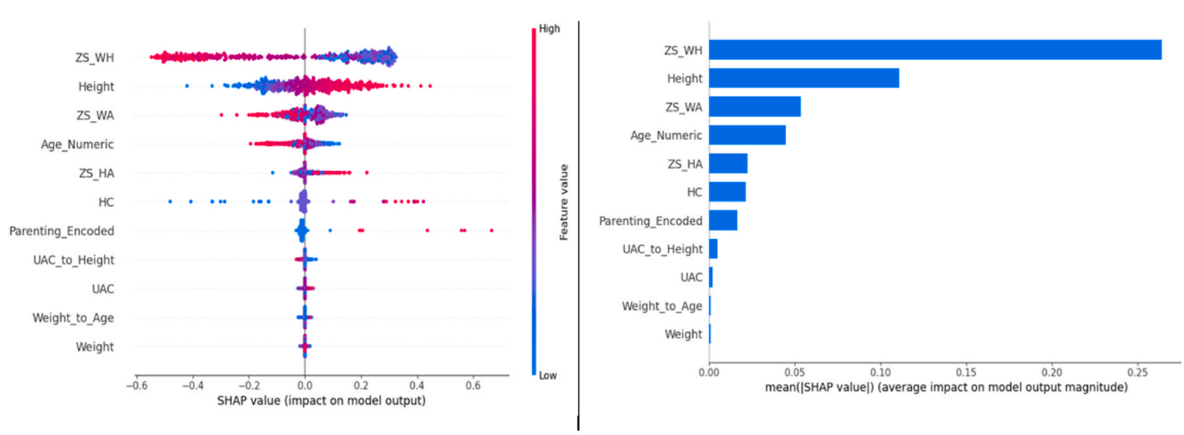

Fig. 9. SHAP summary plot for class 'normal'.

i.e., Positive SHAP values push the model toward predicting the 'Normal' class, while Negative SHAP values reduce the likelihood of the 'Normal' class. This result also confirms the consistency with Domain Knowledge, i.e., the importance of Z-scores (WHO standards for nutritional assessment) aligns with established health metrics, validating the model's reliance on scientifically grounded indicators.

The SHAP summary plot for the class 'Overnutrition' as depicted in Fig. 10 provides insights into how features contribute to the model's prediction. For feature Importance, ZS_WH is the most influential feature for predicting overnutrition, indicating that weight relative to height is a critical indicator in classifying this nutritional status. Other important features include Height, ZS_WA and ZS_HA which align with established anthropometric measures for assessing overnutrition. Furthermore, higher values of **ZS_WH** (red) have a strong positive impact on predicting overnutrition, whereas lower values (blue) reduce the likelihood of this classification. For features like Height and ZS_WA, higher values contribute positively to the prediction of overnutrition, which is consistent with their relationship to excess weight. The alignment of feature importance with domain knowledge confirms the reliability of the model and its potential for deployment in real-world health applications.

The SHAP summary plot for the class 'Undernutrition' as depicted in Fig. 11 provides insights into how features contribute to the model's predictions for this classification. For feature importance, Height emerges as the most critical feature for predicting undernutrition, highlighting its significant role in identifying children with stunted growth (stunting). Other key features include ZS_WH, ZS_WA and ZS_HA, which align with widely used anthropometric measures in assessing malnutrition. The dominance of height and Z-scores (ZS_WH, ZS_WA, ZS_HA) reflects their established importance in identifying undernutrition, particularly stunting and wasting, as per WHO standards.

In addition, the differing feature importance patterns observed in Figs. 10 and 11 reflect the distinct anthropometric characteristics associated with overnutrition and undernutrition cases. In the overnutrition scenario, ZS_WH is the most influential feature, as it directly measures disproportionate weight relative to height, a key indicator of overweight or obesity. In contrast, Height emerges as the most critical feature in the undernutrition case, reflecting its strong relationship with chronic malnutrition (stunting), which is characterized by restricted linear growth over time. This distinction aligns with WHO growth standards, where ZS_WH is commonly used for identifying acute overweight, while Height and Height-for-Age indicators are more relevant for detecting stunting. It confirms that the model is not only accurate but also interprets features in a manner consistent with established clinical guidelines.

4.6. Discussion on experimental outcomes

The performance of CNN and LSTM models was evaluated under three distinct scenarios: extensive data preprocessing, grid search parameter optimization, and AHM integration, each offering unique advantages and challenges. Extensive data preprocessing ensured high data quality by handling missing values, feature selection, scaling, removing correlations, and related pre-processing procedures. Under this scenario, CNN outperformed LSTM with 97.74 % accuracy, while LSTM achieved 96.26 %, highlighting the role of structured spatial data in CNN's predictive power.

Grid search parameter optimization further enhanced performance by fine-tuning hyperparameters, allowing CNN to reach 98.46 % of accuracy, while LSTM performed slightly better with 98.93 % accuracy. This highlights LSTM's ability to adapt to optimized configurations. However, the exhaustive nature of grid search is computationally expensive and time-intensive, making it less scalable for real-time healthcare applications.

The most substantial performance improvement was observed with MHA, which enabled CNN to achieve 99.08 % accuracy and LSTM reaching 98.91 % accuracy. This introduction of MHA allowed the models to dynamically focus on multiple critical features capturing complex cross-feature dependencies that conventional CNN and LSTM models struggle to exploit. This advancement demonstrates how attention-based mechanisms improve predictive performance in structured anthropometric data. MHA's ability to enhance feature prioritization is particularly evident in its impact on ZS_HA, ZS_WH, and ZS_WA, which are the primary indicators for child nutritional assessment. This validates the role of attention mechanisms in improving model accuracy by enabling dynamic feature weighting.

Furthermore, a comparative analysis of CNN with MHA and LSTM with MHA provides deeper insights into their respective strengths. CNN with MHA slightly outperformed LSTM with MHA across all performance metrics, particularly in recall (99.08 % vs. 98.91 %), indicating that CNN is more effective in correctly identifying children at risk of malnutrition. However, LSTM with MHA remains competitive, achieving higher accuracy than CNN without MHA, suggesting that sequential dependency modeling is beneficial when combined with multie head attention mechanisms. These findings provide empirical evidence supporting the evaluation of whether spatial feature learning (CNN with MHA) or sequential dependency modeling (LSTM with MHA) is superior for nutritional status prediction.

To enhance model transparency, SHAP evaluation further validated the models' predictions by identifying high-impact features and ensuring transparency in decision-making. The results confirmed that ZS_WH, ZS_WA, and height were the most influential features across all scenarios, aligning WHO standards for nutritional assesment. SHAP's

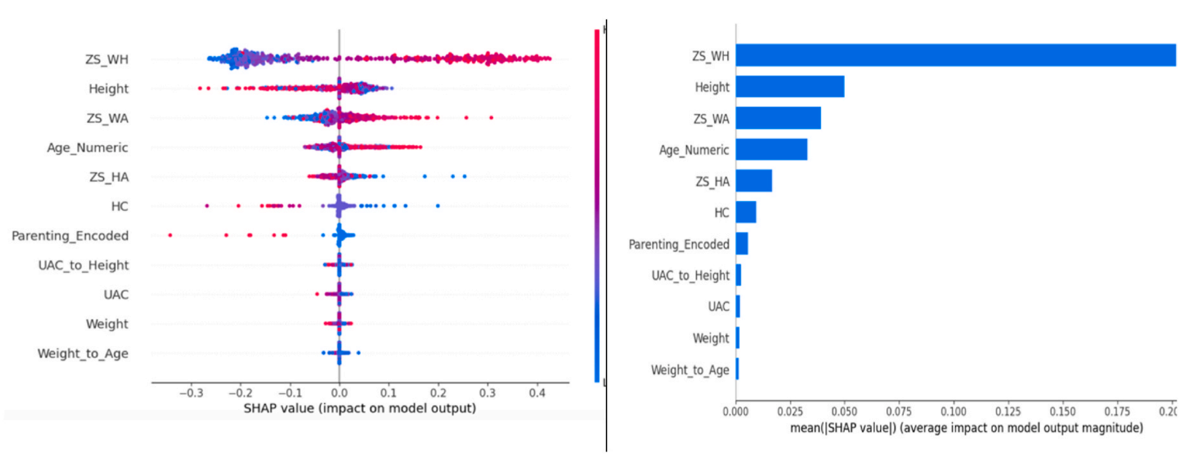

Fig. 10. Shap summary plot for class 'overnutrition'.

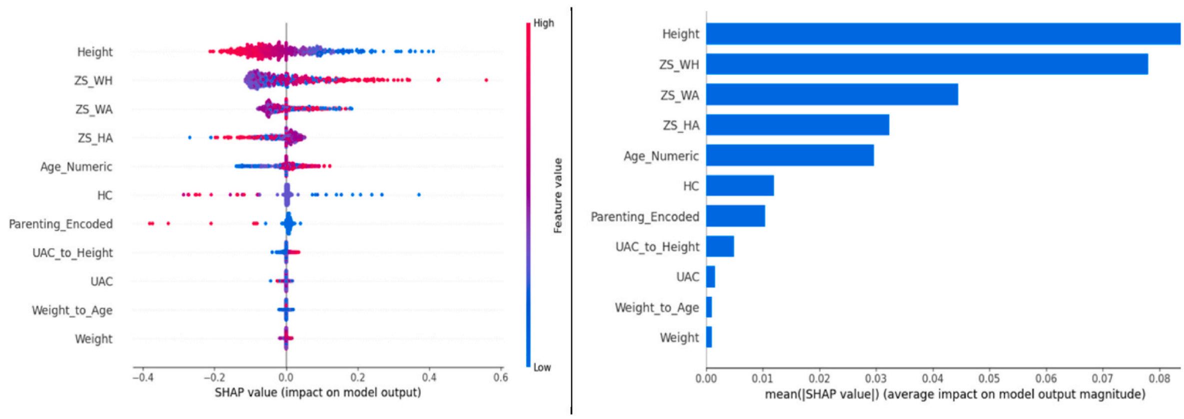

Fig. 11. Shap summary plot for class ‘undernutrition’.

ability to quantify feature contributions highlights the importance of anthropometric measures in predicting health outcomes, validating the model’s reliance on scientifically grounded indicators. For instance, in predicting ‘overnutrition’, SHAP revealed that high ZS_WH values positively influenced the model’s decision, while low values strongly contributed to the prediction of ‘undernutrition’. The transparency provided by SHAP enhances stakeholder trust by making model predictions interpretable and actionable. Health practitioners can use SHAP visualizations to understand how specific features drive predictions, ensuring that interventions are targeted and evidence-based. This level of explainability bridges the gap between black-box ML models and their practical application in healthcare. Additionally, this study presents a practical AI-driven, IoT-integrated nutritional monitoring framework, leveraging real-time data acquisition to minimize errors from manual anthropometric measurements. The ability to combine IoT data with explainable machine learning offers a scalable and efficient approach for early malnutrition detection, demonstrating its potential for real-world healthcare applications.

Unlike to prior studies that relied on traditional machine learning techniques such as Random Forest or SVM, our study integrates CNN and LSTM with multi-head attention mechanisms and XAI techniques. This approach offers both improved accuracy and interpretability, addressing key gaps in existing literature. When benchmarked against the existing literature, CNN with MHA achieved the highest recorded accuracy (99.08 %), demonstrating its superiority in nutritional status prediction. These results underscore the effectiveness of attention-enhanced CNN and LSTM architectures, validating their use for real-world healthcare applications. Furthermore, the integration of XAI tools like SHAP enhances stakeholder trust by providing clinically interpretable results, ensuring that the proposed framework is both scientifically rigorous and practically deployable.

5. Hypotheses on long term-term effects

While the proposed framework effectively predicts children’s nutritional status, its potential long-term implications require further exploration. Although this study primarily focuses on short-term predictive accuracy and explainability, evidence from existing literature indicates that early childhood nutritional status significantly correlates with long-term health outcomes, including cognitive development [67], chronic disease risks [68], and overall well-being [69]. The ability to accurately identify at-risk children during early developmental stages provides an opportunity for targeted interventions that could mitigate adverse health effects in later life.

5.1. Short-term predictions and long-term implications

The predictive capabilities of the proposed model, particularly those leveraging z-scores and anthropometric indicators, align with established WHO growth monitoring guidelines. Prior studies indicated that children classified as undernutrition using z-score thresholds are more likely to experience developmental delays and increased susceptibility to metabolic disorders [70]. Conversely, early indicators of overnutrition are strongly associated with an increased risk of obesity, insulin resistance, and cardiovascular diseases later in life. The integration of multi-head attention within CNN and LSTM models enhances the prioritization of critical anthropometric features, improving the reliability of early nutritional risk assessments. This reinforces the need for proactive health interventions that can mitigate long-term health risks.

5.2. Correlations with longitudinal trends

To assess the long-term significance of the model’s predictions, several studies tracking childhood nutrition and adult health have been carried out. Research has shown that children identified with low height-for-age (stunting) often exhibit reduced cognitive performance in later life. As reported in Ref. [71], stunting and reduced stature serve as key intermediaries linking early-life malnutrition to poor academic performance and economic disadvantages. A decrease of one standard deviation (SD) in ZS_HA is associated with a 0.6-year reduction in schooling, a higher probability of secondary school dropout, and lower cognitive performance. Stunted children score approximately 0.56–0.8 SD lower on cognitive tests, complete two fewer years of schooling, and experience an average school entry delay of 0.4 years. Similarly, early-life overnutrition correlates with an increased risk of diabetes and cardiovascular diseases in adulthood [72]. By establishing quantifiable relationships between early nutrition indicators and future health risks, this study highlights the importance of integrating AI-based monitoring into public health strategies.

5.3. Framework’s contribution to long-term monitoring

The integration of AI-driven prediction within routine health assessments has the potential to revolutionize longitudinal tracking of at-risk children. By systematically analyzing growth patterns, z-score deviations, and nutritional imbalances, the proposed framework can support early interventions aimed at preventing long-term health complications. Future studies could validate these predictions by implementing follow-up monitoring programs, comparing short-term AI-driven classifications with real-world health trajectories over multiple years. Additionally, the integration of IoT-enabled real-time anthropometric data collection will further enhance the framework’s predictive accuracy and applicability in public health settings. By addressing these aspects in future work, this study contributes not only to immediate nutritional status assessment but also to a broader understanding of its long-term implications for child health policy and intervention planning.

6. Conclusion

This study demonstrated the effectiveness of integrating MHA mechanisms with CNN and LSTM models for nutritional status prediction, leveraging data preprocessing and grid search optimization to enhance performance. Although the preprocessing techniques used in this study are not novel individually, their domain-specific application, particularly in support of attention-based learning over structured anthropometric data, adds methodological value and strengthens the reliability of the resulting predictions. CNN-MHA achieved 99.08 % accuracy, while LSTM-MHA reached 98.91 % accuracy, indicating that spatial feature learning with attention is more effective than sequential dependency modeling for structured anthropometric tabular data. Additionally, SHAP analysis validated the key predictive features (ZS_WH, ZS_WA, and height), aligning with WHO standards, ensuring model interpretability and reliability for healthcare applications.

The findings highlight the significance of attention mechanisms in structured anthropometric data modeling, emphasizing their role in enhancing both accuracy and explainability. Future research should explore multi-modal data integration (e.g., dietary intake, physical activity) and optimize computational efficiency to enable real-time deployment in IoT-based health monitoring systems. Additionally, validating models across diverse populations and expanding explainable AI techniques beyond SHAP could further improve transparency and generalizability. By addressing these key areas, future studies can contribute to the advancement of precision nutrition and data-driven public health policies, ensuring that AI-driven nutritional assessments are interpretable, scalable, and actionable for real-world implementation.

CRediT authorship contribution statement

Heru Agus Santoso: Writing – original draft, Project administration, Methodology, Conceptualization. **Nur Setiawati Dewi:** Writing – review & editing, Investigation, Data curation. **Su-Cheng Haw:** Writing – review & editing, Validation, Supervision. **Arga Dwi Pambudi:** Software, Resources. **Sari Ayu Wulandari:** Visualization, Formal analysis.

Ethics statement

This research presents an accurate account of the work performed, all data presented are accurate and methodologies detailed enough to permit others to replicate the work.

This manuscript represents entirely original works and or if work and/or words of others have been used, that this has been appropriately cited or quoted and permission has been obtained where necessary.

This material has not been published in whole or in part elsewhere.

The manuscript is not currently being considered for publication in another journal.

That generative AI and AI-assisted technologies have not been utilized in the writing process or if used, disclosed in the manuscript the use of AI and AI-assisted technologies and a statement will appear in the published work.

That generative AI and AI-assisted technologies have not been used to create or alter images unless specifically used as part of the research design where such use must be described in a reproducible manner in the methods section.

All authors have been personally and actively involved in substantive work leading to the manuscript and will hold themselves jointly and individually responsible for its content.

For ethical clearance, this research was granted by the Universitas Dian Nuswantoro Ethics Committee (002351/UNIVERSITAS DIAN NUSWANTORO/2024) based on WHO 2011 Standard and Guidance Part III, as the study involved human participants.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX

Table 4
Dataset Feature Name and Description

Feature Name	Description
Weight (kg)	Child's weight measured using Load Cell Sensors
Height (cm)	Child's height measured using Rotary Encoder Sensors
UAC	Upper Arm Circumference
HC	Head Circumference
ZS_WA	Weight-for-Age Z-score
ZS_HA	Height-for-Age Z-score
ZS_WH	Weight-for-Height Z-score
UAH	UAC_to_Height
WA	Weight_to_Age
Age	Age of the child at the time of measurement
Parenting Practice	Categorical variable representing feeding/care practices
Nutritional Status Labels	One-Hot Encoded: Normal, Under Nutrition, Overnutrition
IoT Sensor Data Timestamp	Time of data acquisition from IoT-based anthropometric measurement tools
Demographic Data Source	Retrieved demographic information via API from health department records

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ibmed.2025.100255.

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